Course Outline for CSE-143

Part A

1. Course Code: CSE-143

2. Course Title: Object Oriented Programming

3. Course Type: Core Course

4. Level/ Term: Level: 1 Term: II

5. Academic Session: 2020-21

6. Course Teacher: Ashim Dey, Assistant Professor, Dept. of CSE, CUET
Md. Billal Hossain, Assistant Professor, Dept. of CSE, CUET

7. Prerequisite(s): None

8. Credits: 3

9. Contact Hours: 3 lectures of 50 minutes per week

10. Total Marks: 300

11. Rationale of the Course:

This course introduces advanced programming skills and focuses on the core concepts of object-oriented programming and design using a high-level language, C++. The course focuses on the understanding and practical mastery of object-oriented concepts such as classes, objects, data abstraction, methods, method overloading, inheritance and polymorphism. Detailed knowledge of class and object along with Standard Template Library will enable students to implement complex programs systematically and efficiently. This is a required course for all the students enrolling B. Sc. Engg. in CSE program. The catalogue description of the course is

Course Content:

Concepts of object oriented programming, Classes, Friend functions: Objects, isomorphism, polymorphism, inheritance, parameterized constructors, multiple inheritance, passing object to functions, arrays of objects, pointer to objects.

Function and operator overloading, overloading constructor functions, references, virtual functions, Exception Handling, Template functions and classes, Streams, Dynamic allocation, Static class members, Multi-threaded programming.

12. Course Objectives:

a) To study the object oriented programming (OOP) principles, tokens, expressions, control structures and functions.
b) To introduce the classes, objects, constructors and destructors.
c) To cover the operator overloading, inheritance and polymorphism concepts of OOP.
d) To get familiar with memory management, multithreading, managing errors and exceptions and I/O Streams in OOP.
e) To use a programming approach such as C++ established the above mentioned objectives.

13. Course Learning Outcomes (CLOs) and Mapping of CLOs with Program Learning Outcomes (PLOs)

a) CLOs

	Course Learning Outcomes (CLOs)	**Blooms Level (Optional)**
CLO1	Identify the relative merits of C++ as an object oriented programming language while solving problems	C1, C3, P1
CLO2	Apply the major object-oriented concepts such as encapsulation, inheritance and polymorphism to implement object oriented programs in C++	C3
CLO3	Design solutions using advanced features of C++ specifically stream I/O, templates and multithread	C5, P7

b) Mapping of CLO with PLO

No.	**(CLOs)**	**PLO1**	**PLO2**	**PLO3**	**PLO4**	**PLO5**	**PLO6**	**PLO7**	**PLO8**	**PLO9**	**PLO10**	**PLO11**	**PLO12**
1	CLO1	X											
2	CLO2			X									
3	CLO3			X									

Part B

14. Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs

	Topic	Teaching-Learning Methodology	Assessment Method	Corresponding CLOs
Week-01	Introduction to Object-Oriented Programming and overview of C++ basics	• Lecture	Class Test Final Exam	CLO-1
Week -02	Managing input and output	• Lecture	Class Test Final Exam	CLO-1
Week -03	Functions	• Lecture • Assignments	Class Test Final Exam Assignments	CLO-1
Week -04	Concept of Class and Object	• Lecture	Class Test Final Exam	CLO-1
Week -05	Constructors and Destructors	• Lecture • Assignments	Class Test Final Exam Assignments	CLO-1
Week -06	Operator Overloading	• Lecture • Assignments	Class Test Final Exam Assignments	CLO-2
Week -07	Inheritance	• Lecture • Assignments	Class Test Final Exam Assignments	CLO-2

Week -08	Friend Function and Pointers	• Lecture	Class Test Final Exam	CLO-3
Week -09	Polymorphism	• Lecture • Assignments	Class Test Final Exam Assignments	CLO-3
Week -10	File handling and I/O operations	• Lecture • Assignments	Class Test Final Exam Assignments	CLO-3
Week -11	Templates	• Lecture • Assignments	Class Test Final Exam Assignments	CLO-3
Week -12	Exception handling	• Lecture	Final Exam	CLO-2
Week -13	Review and makeup classes	• Lecture	Final Exam	CLO-2

Part C

15. Assessment and Evaluation

l) Assessment Strategy

Class participation and attendance	10%
Class tests / Class assessment	20%
Term Final Examination (3 hours duration)	70%
Total	100%

2) Marks distribution:

a) Continuous Assessment: 30%
b) Summative: 70%
c) Make-up Procedures:
- Feedback on continuous assessment is given to the students immediately after the test.
- The minimum number of class-test/assignments are (n+1) with best n will be counted (here, n is number of credit). Based on the students' feedback additional class-test/assignment may be taken by the course teacher

Part D

16. Learning Materials

1) Recommended Readings

- Book-1: Object Oriented Programming with C++
  - E.Balagurusamy (Tata McGraw Hill Education).
- Book-2: C++: The Complete Reference
  - Herbert Schildt (Mcgraw-Hill Osborne Media).

2) Others

Handout/lecture material provided by the course teacher